An energy budget

antocciphysics | TEACHER KEY | CPA

Learning target: Connect work, motion, heating and an honest model conclusion.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 18; 3 course days later.*
3 J remains unaccounted. Seek another storage/transfer or check measurements.

02. Explain a worked example (worked)
Useful fraction 60%; outgoing branches total input.

03. Look and explain (model)
100=60+25+15; balanced.

04. Build the solution (practice)
Initial 16 J; ideal 24 J; with thermal transfer final 19 J.

05. Use the idea (reason)
Need partition, entry losses, ejecta, air blast, deformation, radiation and coupling; claim is unsupported.

06. Choose a method (mixed) *Review: Unit 4, Day 9; 12 course days later.*
$\Delta U = +20$ J; gained. Less negative means greater energy.

07. Exit check (exit)
10 J other; 75%useful.

08. STRETCH: Apply it another way (stretch)
E.g. thermal 15, other 10; or thermal 20, other 5 J. Both balance but are hypotheses.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 18; 3 course days later.*
Energy=3.75 J; average opposing force≈5.357 N.

02. Explain a worked example (worked)
Useful fraction 60%; outgoing branches total input.

03. Represent and interpret (model)
100=60+25+15; balanced.

04. Solve and justify (practice)
Initial $+2\times10^{6}$ J; after -1×10^{6} J. Negative energy is bound in ideal isolated gravity model; positive permits unbound trajectory with conditions.

05. Analyze the reasoning (reason)
Need partition, entry losses, ejecta, air blast, deformation, radiation and coupling; claim is unsupported.

06. Choose a method (mixed) *Review: Unit 4, Day 9; 12 course days later.*
U_i=-48 J; U_f=-24 J; ΔU=+24 J; gravity work=-24 J.

07. Exit check (exit)
Initial 60 J; K 48 J; $v=\sqrt{48}=6.93$ m/s.

08. STRETCH: Apply it another way (stretch)
Input 3000 J; useful 1200 J; efficiency 40%; remaining 1800 J.